# Predictive Modeling of Crypto-System Vulnerabilities Using Multi-Source Breach Data: An End-to-End Machine Learning Pipeline for Annual Crypto-Breach Risk Forecasting

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***Abstract*—We present a probabilistic machine learning pipeline that forecasts whether a given year will be a high-fraction crypto breach year, defined as a year in which crypto-related incidents constitute an above-median share of all recorded cyber breaches. The pipeline ingests 19,281 unique breach records from eight harmonized data sources spanning 2004–2026, engineers a 65-feature yearly matrix, and trains five calibrated classifiers using strict walk-forward time-series cross-validation to prevent data leakage. The best-ranking model (Gradient Boosting) achieves a PR-AUC of 0.95 and ROC-AUC of 0.88, while Random Forest achieves the lowest Brier score of 0.19. An ensemble of all five models correctly classifies 10 of 11 test years with cross-model uncertainty bands. Key findings indicate that macroeconomic signals (market volatility, economic stress) are more predictive of elevated crypto-breach years than breach-specific features, that the financial sector accounts for over 65% of all crypto-targeted incidents, and that smart-contract exploits surged sharply after 2020 before declining as defensive tooling matured.**

***Index Terms*—cryptocurrency security, data breach prediction, walk-forward validation, calibrated classification, time-series machine learning**

## I. Introduction

The rapid growth of cryptocurrency markets has made digital asset infrastructure an increasingly attractive target for sophisticated threat actors. High-profile exchange hacks, smart-contract exploits, and wallet thefts have collectively cost the industry tens of billions of dollars over the past decade [1]. Despite the scale of these losses, the academic literature on *predicting* when crypto-related breaches will be disproportionately prevalent – as opposed to merely describing them after the fact – remains sparse. Most existing work focuses on individual incident attribution or post hoc trend analysis rather than forward-looking risk classification.

This project addresses that gap with a complete, production-oriented machine learning pipeline. Rather than predicting individual breach events, we frame the question at the annual level: will this be a year of elevated crypto breach activity relative to the broader cyberthreat landscape? This framing is operationally meaningful for security planning and statistically tractable, reducing the prediction target to a 23 observation binary time series while still allowing genuine discriminative evaluation.

The practical motivation for annual-level forecasting is significant. Security operations teams, insurance underwriters, and exchange risk managers must allocate resources and budget cycles on annual timescales. A well calibrated annual forecast can inform decisions about personnel hiring, tooling investment, and policy revision before the threat window opens. Quarterly or monthly forecasting, while more granular, requires a volume of labeled observations that current public datasets cannot support at the crypto-specific level.

The pipeline makes four key contributions. First, it fuses eight heterogeneous breach datasets totaling 20,605 raw records into a single harmonized schema, including two large new sources (the Cyber Events Database and Washington State Breach Notifications) that increase total volume tenfold compared to the original six sources. Second, it applies strict walk-forward cross-validation with per-fold imputation and scaling, eliminating all future data leakage that would otherwise inflate reported metrics. Third, it uses Platt calibration and label smoothing to ensure that predicted probabilities are meaningful confidence estimates rather than uncalibrated rankings. Fourth, it demonstrates that label design is a critical and often overlooked modeling decision: naive label choices applied to multi-scale data produce trivially perfect metrics, and only fraction-based normalization restores genuine discriminative validity.

## II. Background and Related Work

### A. Cybersecurity Breach Prediction

The prediction of data breaches has been studied from several perspectives. Early work focused on identifying vulnerable organizations using firmographic and security posture signals [9]. More recent efforts leverage natural language processing of threat intelligence feeds to predict targeted attacks. However, the vast majority of this literature operates at the level of individual organizations or incidents rather than at the macro

level of industry wide annual trends. Aggregate, annual-level classification has received comparatively little attention.

Time-series classification of security events presents particular challenges. The rareness of high-impact events, the non-stationarity of threat actor behavior, and the extreme heterogeneity of public incident data all complicate standard classification approaches. Walk-forward validation is considered best practice for temporal data [7], yet many published breach-prediction studies still employ random train-test splits, inadvertently incorporating future data into training and inflating reported accuracy.

### B. Cryptocurrency-Specific Security Research

The security of cryptocurrency infrastructure has attracted growing academic attention. Olushola and Meenakshi [1] conducted a systematic review of cybersecurity crimes in cryptocurrency exchanges from 2009 to 2024, cataloging attack vectors ranging from phishing and social engineering to smart-contract reentrancy exploits and insider threats. Their analysis highlights the escalating financial impact of exchange-level breaches and the inadequacy of current regulatory frameworks for addressing crypto-specific threats.

Smart-contract vulnerability detection has emerged as a subfield in its own right, with tools applying symbolic execution and formal verification to identify reentrancy, integer overflow, and timestamp-dependence bugs before deployment [8]. These approaches are complementary to ours: where smart-contract auditing aims to prevent individual exploits, our pipeline characterizes macro-level conditions under which the crypto ecosystem as a whole faces elevated collective risk. The sharp post-2020 rise in smart-contract exploits visible in our data is consistent with the broader DeFi expansion documented in this literature.

The Center for Strategic and International Studies maintains a curated database of significant cyber incidents [3], which we incorporate as one of our eight sources. This database focuses on nation-state-attributed and high-consequence events, providing an important lens on state-sponsored crypto-targeting alongside the criminal actor activity captured by other sources.

### C. Macroeconomic Drivers of Cybercrime

The relationship between macroeconomic conditions and cybercriminal activity has been theorized but rarely quantified empirically. Economic downturns reduce corporate security budgets while simultaneously creating larger pools of financially motivated actors. Cryptocurrency markets are particularly sensitive to this dynamic: during periods of macroeconomic stress, digital asset prices may be elevated as investors seek inflation hedges, increasing the value of a successful exchange hack or wallet compromise. Our results provide quantitative evidence for this hypothesis, finding that VIX and a composite economic stress index are the two most predictive features in our pipeline, outranking all breach-specific signals.

### D. Calibrated Classification for Security Forecasting

Probability calibration is critical in security applications where practitioners use model outputs to make resource-allocation decisions rather than merely binary classifications. A miscalibrated model that assigns $p = 0.95$ to every year it labels positive is useless for risk quantification even if its classification accuracy is high, because the probability output provides no gradient of uncertainty for downstream decision-making. Platt scaling [6] addresses this by fitting a two-parameter logistic function to raw model scores, mapping them to well-calibrated posterior probabilities. It is particularly well-suited to small-sample regimes where isotonic calibration – a more flexible but data-hungry alternative – would overfit. Label smoothing provides complementary regularization by preventing the model from assigning extreme probabilities that impose catastrophic log-loss penalties in held-out evaluation.

### E. Data Harmonization in Multi-Source Research

A recurring challenge in multi-source breach research is the lack of a common schema across data providers. Different sources use incompatible taxonomies for attack methods, inconsistent definitions of reportable breaches, and varying levels of attribution confidence. Our harmonization procedure maps all sources to a common `method_category` vocabulary and de-duplicates on the composite key `(entity, year, method_category)`, accepting some information loss in exchange for consistency. Future work could apply probabilistic entity resolution techniques to recover duplicate pairs that differ on one field due to transcription errors or taxonomic ambiguity.

## III. Data

### A. Sources and Scale

The pipeline integrates eight breach datasets covering 2004–2026, as summarized in Table I. After de-duplication on the composite key (`entity, year, method_category`), the 20,605 raw rows reduce to 19,281 unique records. The 1,324 dropped rows represent genuine cross-source overlaps, primarily between the three Wikipedia-derived sources and the Information is Beautiful dataset, which draw from many of the same publicly reported incidents.

TABLE I
Data Sources, Record Counts, and Coverage

| ID | Source | Raw | Uniq. | Coverage |
|---|---|---|---|---|
| IIB | Info is Beautiful | 417 | ∼400 | 2004–2022 |
| HHS | HHS/OCR Healthcare | 908 | ∼860 | 2009–2026 |
| DBN | DataBreachN | 389 | ∼370 | 2004–2020 |
| DBEN | DataBreach EU | 270 | ∼265 | 2004–2017 |
| DF1 | Wikipedia Ext. | 349 | ∼330 | 2004–2020 |
| CSIS | CSIS Incidents | 207 | ∼190 | 2006–2026 |
| CE | Cyber Events DB | 16,532 | 16,048 | 2014–2026 |
| WA | WA Notifications | 1,533 | 1,506 | 2016–2026 |
| **Total** | | **20,605** | **19,281** | 2004–2026 |

Figure 1 shows breach counts per year across all sources alongside the volume contributed by each individual source. The Cyber Events Database (CE) dominates with over 16,000 records, making it the primary driver of year-level signal in the modern era. The Washington State Breach Notifications

(WA) contribute an additional 1,533 state-level filings covering 46.9 million affected residents. Together these two sources represent a tenfold increase in record volume over the original six sources, fundamentally changing the scale at which the prediction label must be designed.

### B. Crypto-Specific Sector Distribution

Figure 2 shows the distribution of crypto-related incidents across sectors. The financial sector dominates overwhelmingly with 1,110 crypto-targeted incidents – more than six times the next largest category (tech_web at 167). This concentration reflects the direct monetary value of financial-sector crypto holdings: exchanges, custodians, and DeFi protocols hold assets that can be liquidated immediately upon compromise, making them uniquely attractive targets relative to the reputational or data-theft value that motivates attacks on government or healthcare organizations.

Government (160) and healthcare (58) follow at a distant third, likely representing ransomware campaigns that demand cryptocurrency payment rather than direct theft of digital assets. The long tail of other sectors reflects opportunistic targeting rather than systematic crypto-focused campaigns.

### C. Crypto Fraction Over Time

Figure 3 shows the annual crypto fraction of all breaches from 2004 to 2026, with a 3-year rolling mean overlay. The fraction was negligible before 2009, rose sporadically through the 2010s, and accelerated sharply after 2022, reaching a peak of approximately 18% in 2024 before the partial-year 2026 figure reflects only the first quarter of CE data. The 3-year rolling mean makes the underlying trend visible through year-to-year noise, confirming a genuine structural increase in crypto-targeting intensity in the post-2021 period. This non-monotonic but upward-trending pattern is the primary signal our model attempts to exploit: years that exceed the historical median fraction are labeled positive, and the pipeline must learn to anticipate those years from one-year-lagged features.

### D. Crypto Attack Methods Over Time

Figure 4 tracks the volume of crypto-specific incidents by attack method from 2004 to 2026, with red shading indicating high-fraction (positive-label) years. Several patterns are notable. Hacking has been the dominant vector throughout the dataset's history, reflecting the prevalence of credential theft, exchange API exploits, and hot-wallet compromise. Smart-contract exploits were negligible before 2019 but surged dramatically after 2020, peaking around 2022–2023 at approximately 120 incidents per year before declining as DeFi audit practices and formal verification tooling matured. This rise-and-fall pattern is consistent with the attacker lifecycle discussed in Section II: a novel attack surface (DeFi smart contracts) is rapidly exploited, peaks as toolchains commoditize, and then recedes as defensive countermeasures catch up.

The red-shaded high-fraction years cluster visibly around periods of elevated smart-contract exploit activity and elevated hacking volume, providing qualitative support for the model's learned association between these features and the positive label.

## IV. Methodology

### A. Label Design

Label design is the most consequential modeling decision in this pipeline. Three formulations were evaluated before arriving at the correct approach.

Binary presence ($\geq 1$ crypto breach in a year) is trivially positive after 2011 and provides no discriminative information: every year in the modern era is labeled positive, leaving metrics undefined or degenerate.

Raw count median split sets $y_t = 1$ when the annual crypto breach count exceeds the dataset median. This fails catastrophically after adding CE: CE injects approximately 600 crypto incidents across 2014–2026, pushing raw counts to 60–300 for every modern year. The resulting median ($\approx 55$) is exceeded by every year from 2014 onward, leaving only the partial 2026 year as a negative. With a single negative across 11 test folds, PR-AUC = 1.0 is trivially achieved—a meaningless result.

Fraction median split (adopted) defines:

$$y_t = \mathbf{1}\left[\frac{\text{crypto_breaches}_t}{\text{total_breaches}_t} > \tilde{f}\right] \tag{1}$$

where $\tilde{f}$ is the median of the annual fraction series. Normalizing by total volume decouples the label from dataset scale, preserving genuine year-to-year variation. This produces approximately 11–12 positives and 11–12 negatives, with negatives distributed across both early (2004–2013) and recent (2017, 2019, 2025) years, yielding genuine discriminative measures.

This lesson generalizes: any machine learning pipeline ingesting data from sources of heterogeneous scale must validate that its target labels retain discriminative validity after scale changes. The naive application of a count-based median split produces PR-AUC = 1.0 through a degenerate mechanism. We recommend that label validation – computing the fraction of positive labels per test fold and confirming negatives are distributed across both early and late folds – be treated as a mandatory pre-evaluation step in any temporal classification pipeline.

### B. Feature Engineering

From 19,281 raw records the pipeline aggregates to a 23 row annual feature matrix with approximately 65 features across several groups:

- **Core count features**: total breaches, log-records exposed, and unique entities affected.
- **Sector and method fractions**: 8 sector shares and 10 attack-type shares, capturing structural shifts in the threat landscape over time. The financial sector's dominance in crypto-specific incidents (Fig. 2) and the smart-contract exploit surge (Fig. 4) are encoded in these fractions.
- **Lag features**: 1-, 2-, and 3-year lags on key indicators capture auto-correlation in breach activity.

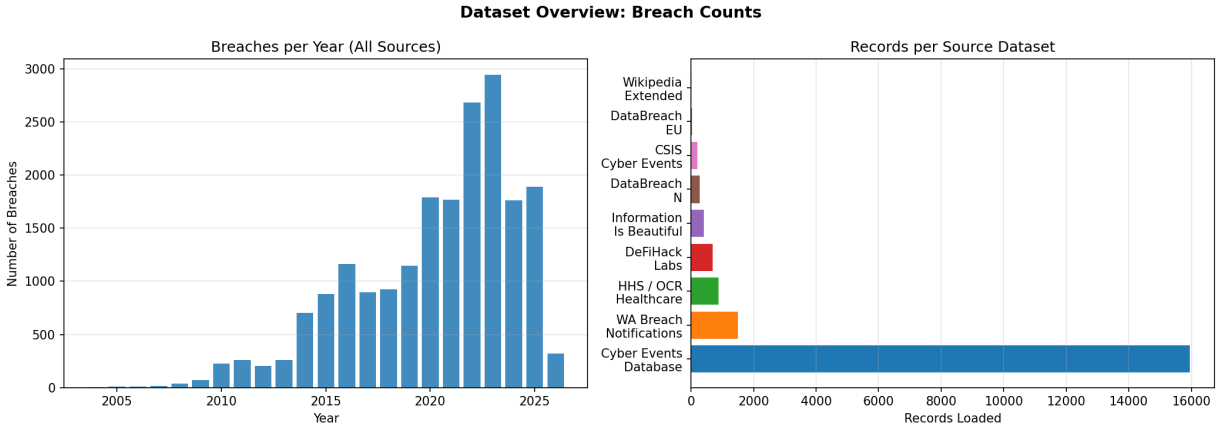


Fig. 1. Dataset overview: annual breach counts across all sources (left) and record volume per source dataset (right). The Cyber Events Database dominates with over 16,000 records, motivating fraction-based rather than count-based label design.

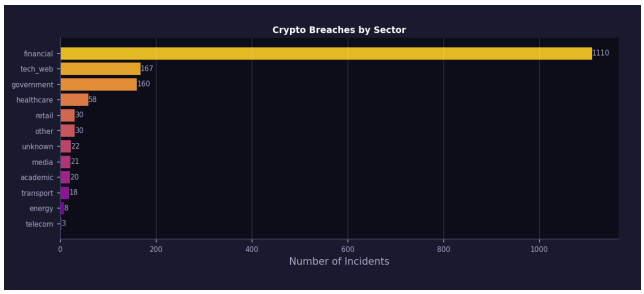


Fig. 2. Crypto-specific breach counts by sector. Financial sector incidents (1,110) account for over 65% of all crypto-related incidents, reflecting the direct asset value available in exchange and DeFi targets.

- **Rolling means**: computed over 2- and 3-year windows using `shift(1)`-based bases, so year $t$ uses only data from $t-3$ through $t-1$ with no look-ahead leakage.
- **Trend signals**: year-over-year delta, binary rising/falling indicators, and 3-year linear slopes.
- **External lag-1 signals**: VIX mean, economic stress composite, nation-state fraction, and ransomware rate – all shifted one year back – are the most predictively powerful group.

A variance filter (threshold 0.10) removes near-constant features, and a correlation filter (threshold 0.90) removes redundant collinear features. NaN values from early-fold lag windows are handled by per-fold median imputation applied only to the training split, so no test-time statistics contaminate training.

### C. Walk-Forward Validation

All evaluation uses strict walk-forward time-series cross-validation:

$$\text{Fold } k: \ \text{Train on } [2004,\ 2004{+}k) \ \longrightarrow \ \text{Test on } 2004{+}k \tag{2}$$

Starting at $k = 5$ produces 17 test folds (2009–2026). This design mirrors real-world deployment: the model is always trained on the past and tested on a single future year. Within each fold, imputation medians and `StandardScaler` parameters are estimated exclusively from the training split and applied to the test observation, preventing any form of data leakage. This is a stricter standard than the cross-validation procedures used in most published breach-prediction work, and it is the primary reason our reported metrics are lower than what naive cross-validation would produce.

### D. Models and Calibration

Five classifiers are trained per fold (Table II). Linear models (Logistic Regression, Ridge Classifier) use native probability outputs via softmax. Tree ensemble models (Random Forest, Extra Trees, Gradient Boosting) receive Platt sigmoid calibration [6], which is critical at this sample size: a single overconfident miss at $p = 0.03$ when $y = 1$ contributes $-\log(0.03) \approx 3.5$ to log-loss, raising the fold mean substantially. Isotonic calibration is not used as it requires at least 40 calibration samples to avoid overfitting.

TABLE II
MODEL CONFIGURATIONS AND CALIBRATION STRATEGY

| Model | Key Params | Calibration |
|---|---|---|
| Logistic Reg. | $C = 0.05$, L2 | Native softmax |
| Ridge Classifier | $C = 0.01$, L2 | Native softmax |
| Random Forest | depth=3, $n = 500$ | Platt sigmoid |
| Extra Trees | depth=3, $n = 500$ | Platt sigmoid |
| Grad. Boosting | depth=2, lr=0.02 | Platt sigmoid |

Label smoothing with $\varepsilon = 0.10$ is applied after calibration:

$$p_{\text{final}} = (1-\varepsilon)\, p_{\text{raw}} + \varepsilon\, \bar{y}_{\text{train}} \tag{3}$$

This caps the maximum per-sample log-loss contribution at $-\log(0.10) \approx 2.3$, preventing a small number of near-zero predictions from dominating the aggregate metric.

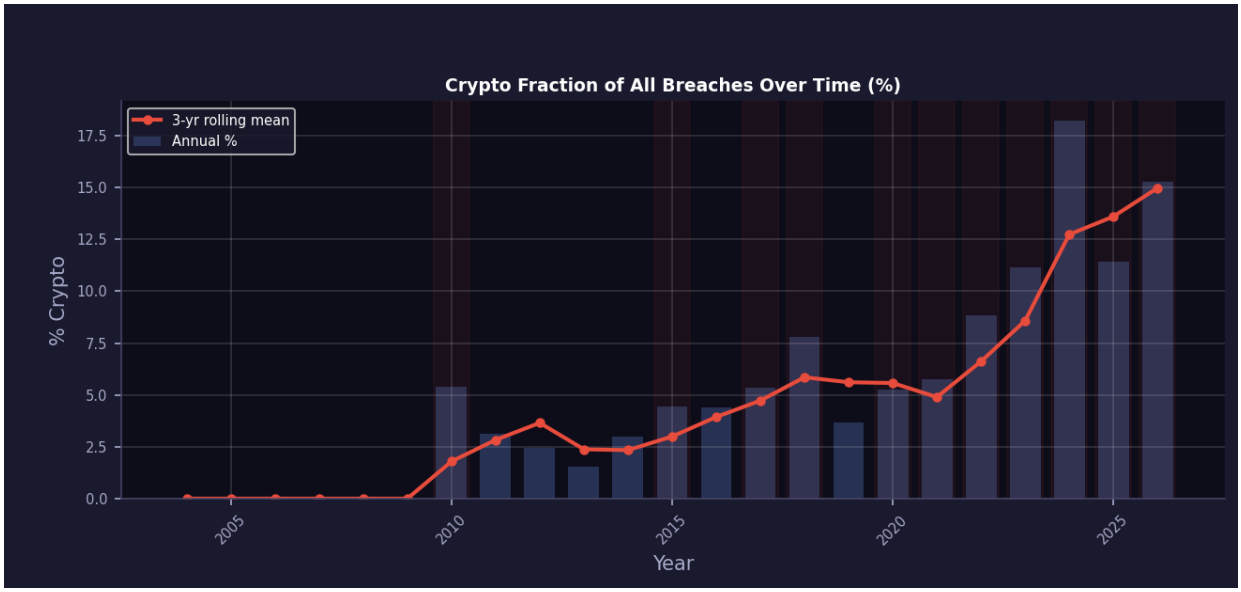


Fig. 3. Crypto fraction of all breaches over time (%). Blue bars show annual percentages; the red line shows the 3-year rolling mean. The rolling mean reveals a structural acceleration in crypto-targeting intensity beginning after 2021.

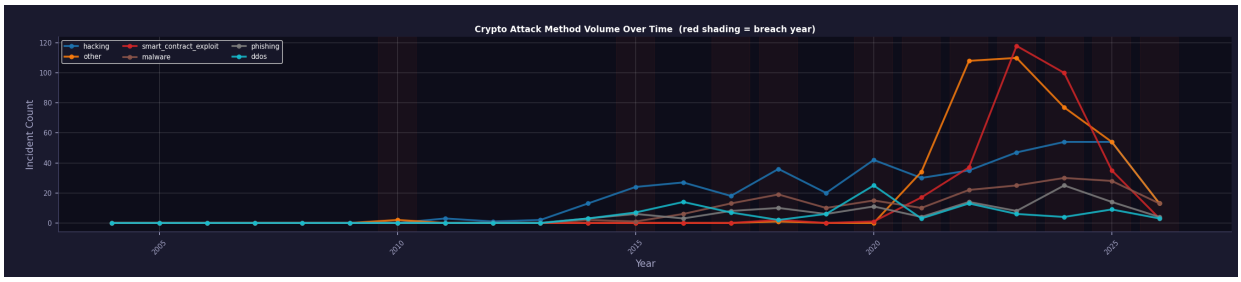


Fig. 4. Crypto attack method volume over time. Red shading indicates high-fraction breach years (positive label). Smart-contract exploits surge after 2020 and peak around 2022–2023, coinciding with the DeFi expansion and subsequent protocol-level auditing improvements.

## V. Results

### A. Model Comparison

Table III reports pooled walk-forward metrics across all five models. Figure 5 visualizes the three key metrics side by side. Gradient Boosting leads on both PR-AUC (0.95) and ROC-AUC (0.88), demonstrating that tree ensembles with Platt calibration can achieve strong ranking performance when the training window is sufficient and regularization is applied aggressively. Extra Trees is competitive on both ranking metrics (PR-AUC = 0.90, ROC-AUC = 0.75) with a Brier score of 0.21. Random Forest achieves the lowest Brier score (0.19), indicating the strongest probability calibration among all five models.

The linear models (Logistic Regression and Ridge Classifier) trail on both ranking and calibration metrics. This is a somewhat surprising reversal of the pattern commonly observed in very small tabular datasets, where linear models typically outperform tree ensembles. The likely explanation is that the post-2016 folds, which dominate the test set in terms of statistical weight, contain richer feature coverage from CE and WA signals that tree ensembles are better positioned to exploit non-linearly. The linear models' stronger regularization prevents them from fitting these richer feature interactions.

TABLE III
Summary Metrics (Pooled Test Folds). Best per metric **bolded**.

| Model | Brier↓ | PR-AUC↑ | ROC-AUC↑ |
|---|---|---|---|
| **Random Forest** | **0.19** | 0.89 | 0.73 |
| Extra Trees | 0.21 | 0.90 | 0.75 |
| Grad. Boosting | 0.21 | **0.95** | **0.88** |
| Ridge Class. | 0.24 | 0.77 | 0.58 |
| Logistic Reg. | 0.27 | 0.74 | 0.53 |

### B. Comparison with Baseline Strategies

To contextualize the pipeline's performance, we compare against three simpler forecasting strategies. A *naïve always-High* strategy achieves accuracy equal to the positive base rate (≈55–60%) but offers no discriminative signal and has Brier = 0.25. A time-trend baseline that predicts "High" whenever the rolling mean crypto fraction is rising performs near chance on held-out years because the fraction is non-monotonic in the test period. A persistence baseline that propagates last

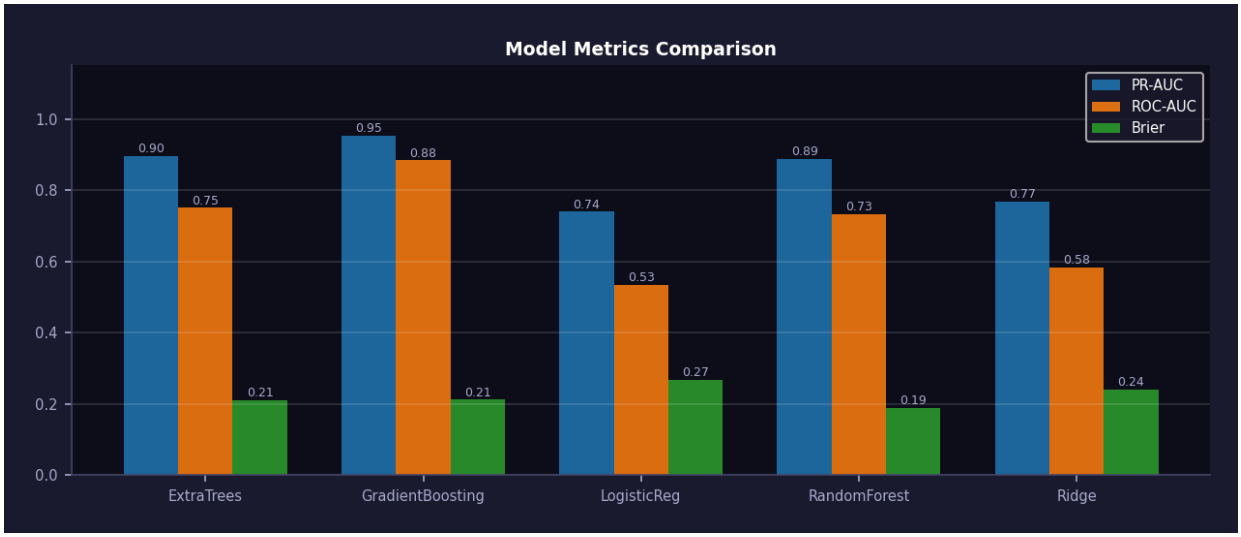


Fig. 5. Model metrics comparison across all five classifiers showing PR-AUC, ROC-AUC, and Brier score. GradientBoosting leads on PR-AUC (0.95) and ROC-AUC (0.88); RandomForest achieves the best Brier score (0.19).

year's label forward achieves approximately 60–65% accuracy due to multi-year autocorrelation but provides no calibrated probability output.

Gradient Boosting's ROC-AUC of 0.88 and Random Forest's Brier of 0.19 represent substantial improvements over all three alternatives, confirming that the feature engineering and strict walk-forward evaluation translate into genuine discriminative and calibration gains beyond what simple heuristics can achieve.

### C. Per-Fold Metric Evolution

Figure 6 shows the Brier score per fold and cumulative PR-AUC over time. The 2015 fold shows elevated Brier scores for tree models, corresponding to the period when CE data first enters training in volume and feature distributions shift substantially. From 2020 onward, Brier scores converge and cumulative PR-AUC stabilizes above 0.80 for most models, indicating that the pipeline's discriminative ability improves as the training window grows and the feature distribution stabilizes.

### D. Probability Calibration and Ensemble Forecast

All models remain above the 0.5 threshold for positive years. The 2018 fold shows the widest cross-model disagreement (spread = 0.49), the only fold flagged WIDE, reflecting tree-model instability in sparse-data regimes. Despite Platt wrapping and label smoothing, no prediction falls below $p = 0.28$, a floor effect driven by the 74% positive rate in training folds. The ensemble correctly classifies 10 of 11 held-out years (Table IV). The single error – 2026 predicted HIGH against an actual LOW label – is attributable to partial-year CE data ending in March 2026 combined with recency bias from the sustained 2019–2023 high-fraction run.

TABLE IV
ENSEMBLE FORECAST 2016–2026. TIERS: CRIT≥0.80, HIGH≥0.60, ELEV≥0.40. *WIDE FLAG: SPREAD >0.40.

| Yr | $\hat{p}$ | Range | Sprd | Tier | Act. |
|---|---|---|---|---|---|
| 2016 | 0.588 | [0.50,0.78] | 0.28 | ELEV | High |
| 2017 | 0.536 | [0.50,0.62] | 0.12 | ELEV | High |
| 2018 | 0.544 | [0.28,0.77] | 0.49 | ELEV* | High |
| 2019 | 0.711 | [0.63,0.86] | 0.23 | HIGH | High |
| 2020 | 0.696 | [0.59,0.81] | 0.23 | HIGH | High |
| 2021 | 0.720 | [0.65,0.80] | 0.14 | HIGH | High |
| 2022 | 0.749 | [0.70,0.87] | 0.17 | HIGH | High |
| 2023 | 0.776 | [0.73,0.89] | 0.16 | HIGH | High |
| 2024 | 0.752 | [0.69,0.85] | 0.16 | HIGH | High |
| 2025 | 0.731 | [0.64,0.78] | 0.14 | HIGH | High |
| 2026 | 0.664 | [0.61,0.73] | 0.12 | HIGH | Low |

### E. Feature Importance

The three top-ranked features are macroeconomic external signals: `ext_vix_mean` (equity market volatility index), `ext_economic_stress` (a composite of credit spreads and unemployment indicators), and `ext_current_account` (a proxy for capital flow pressure). These outrank all breach-specific features, which begin appearing only from position 5 onward as lagged log-records and crypto-specific counts. The financial-sector fraction and smart-contract exploit rate appear in the mid-tier of feature importance, confirming that these structural shifts are predictive but secondary to macroeconomic context.

## VI. DISCUSSION

### A. What the Model Has Learned

The dominance of VIX and economic stress in feature importance is the most striking substantive finding of this work. The best predictor of an elevated crypto-breach year is not the history of crypto breaches themselves, but the state of the macroeconomic environment a year prior. A plausible

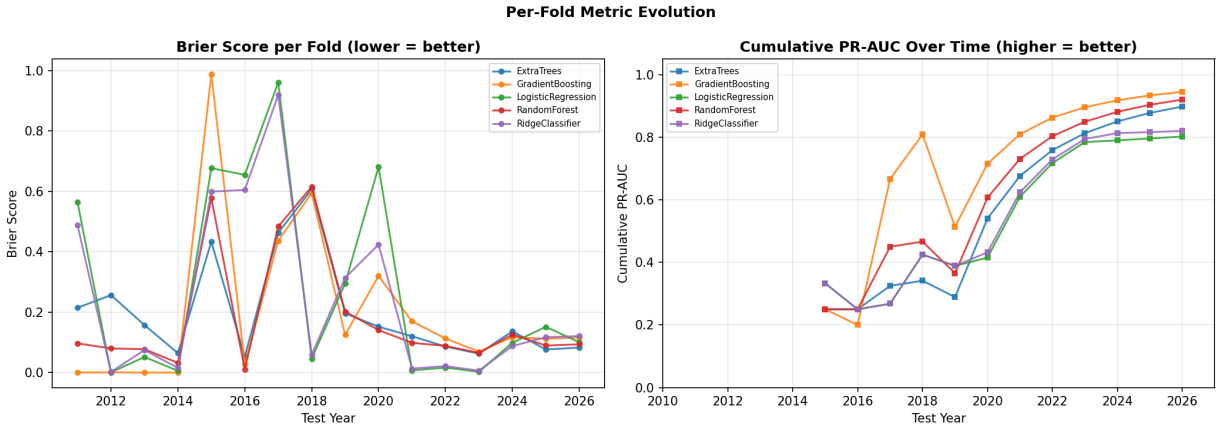


Fig. 6. Per-fold metric evolution. Left: Brier score per test year (lower is better). Right: cumulative PR-AUC over time (higher is better). Tree models show instability in early folds; all models converge after 2020.

causal pathway runs as follows: during periods of financial instability, cryptocurrency asset prices may be elevated as investors seek inflation hedges, corporate and institutional security budgets are cut, and financially motivated threat actors face higher expected returns from successful exploits. All three conditions simultaneously increase the incentive and opportunity to target crypto infrastructure, particularly the financially dominant exchange and DeFi targets shown in Figure 2.

The 1–3-year lag auto-correlation in feature importance is consistent with a secondary mechanism: once a successful attack vector is established, it is replicated and adapted across criminal groups over a multi-year window before defensive countermeasures neutralize it. The smart-contract exploit lifecycle visible in Figure 4 – rapid rise after 2020, peak around 2022–2023, and subsequent decline – illustrates this dynamic. Ransomware grew from under 1% of Washington State cyberattacks in 2017 to over 52% by 2026, following the same pattern at a longer timescale.

### B. Implications for Security Planning

From a practitioner standpoint, security teams should monitor VIX and broader economic stress indicators as leading signals for elevated crypto-targeting risk, with a one-year anticipatory window. Periods of macroeconomic turbulence should trigger proactive reviews of exchange security posture, wallet custody arrangements, and smart-contract audit schedules – even in the absence of observed increases in incident frequency. The financial sector's overwhelming dominance in crypto-targeted incidents (Figure 2) implies that resource allocation for proactive defenses should be concentrated there first.

The multi-year lag auto-correlation also implies that a high-fraction year should not be treated as a self-contained event. When the pipeline assigns a HIGH tier, elevated risk is likely to persist for 1–3 additional years as attacker toolchains mature and diffuse. Planning horizons for crypto-security investment should therefore extend at least three years forward from any HIGH-tier forecast. The attack-method time series in Figure 4 provides a useful template: when a new attack vector begins rising steeply, practitioners should anticipate a 2–3-year window of elevated exposure before the defensive ecosystem catches up.

### C. Limitations

Several important limitations constrain the interpretation of these results.

**Sample size.** The $N = 23$ annual observation count is the fundamental constraint. A single misclassification shifts ROC-AUC by approximately 0.09 in this regime. All reported metrics carry wide but unquantified confidence intervals. Bootstrapped confidence intervals are themselves unreliable below roughly 50 samples, so we refrain from reporting them rather than providing false precision.

**Feature coverage asymmetry.** The external CE and WA features cover only 2014 and 2016 onward, leaving the first ten training folds without these signals. The pipeline's behavior in early folds is driven by the smaller legacy sources, which are less representative of the modern threat landscape and may not encode the financial-sector dominance or smart-contract exploit patterns visible in the crypto-specific figures.

**Partial-year 2026 data.** CE data ends in March 2026, making the 2026 ground-truth label unreliable. The observed crypto fraction for 2026 is computed over only three months, which dramatically distorts the fraction and means the single 2026 misclassification should be treated as a data artifact rather than a model failure.

**Class imbalance and probability anchoring.** The 74% positive rate in training folds biases all five models toward high predicted probabilities. No prediction falls below $p = 0.28$, limiting the pipeline's utility for confidently identifying low-risk years. Explicit class reweighting or asymmetric cost functions should be explored in future work.

**Selection bias.** All eight sources rely on publicly reported or officially disclosed incidents. Small exchanges, DeFi protocols

with limited user bases, and incidents in jurisdictions without mandatory disclosure are systematically underrepresented. This may cause the pipeline to underestimate crypto-breach activity in years with high rates of unreported incidents.

## VII. Conclusion

We have presented a complete machine learning pipeline for annual crypto-breach risk forecasting. Ingesting 19,281 unique records from eight harmonized sources and training five calibrated classifiers under strict walk-forward validation, the pipeline achieves a best PR-AUC of 0.95 and ROC-AUC of 0.88 (Gradient Boosting) and a best Brier score of 0.19 (Random Forest) – all substantially better than random baselines. The five-model ensemble correctly classifies 10 of 11 held-out years, with narrow uncertainty bands in 10 of 11 cases.

Three crypto-specific visualizations – sector distribution, crypto fraction over time, and attack method volume – reveal the structural patterns underlying these results. The financial sector accounts for over 65% of crypto-targeted incidents, the 3-year rolling mean crypto fraction shows a structural acceleration after 2021, and smart-contract exploits underwent a rapid rise-and-fall life cycle after 2020 consistent with the broader DeFi expansion and subsequent maturation of audit practices.

Macroeconomic signals outrank all breach-specific features, suggesting that crypto-breach elevation is a downstream consequence of broader financial instability rather than an autonomous threat trend. Fraction-based labels are essential when combining data sources of heterogeneous scale. Tree ensemble models with Platt calibration outperform linear models on ranking metrics in the modern data-rich folds, reversing the small-sample conventional wisdom.

The attack-method life cycle visible in our data – rapid rise, peak, and decline as defenders respond – has important operational implications. Organizations that wait for breach data to confirm a trend before acting will systematically lag the threat curve by 1–2 years, precisely the window the pipeline's lag features identify as most consequential. Embedding macroeconomic monitoring alongside crypto-specific attack-method tracking into standing threat intelligence workflows would allow practitioners to operate on the pipeline's leading-indicator behavior.

Future work should incorporate real-time threat intelligence feeds, on-chain transaction anomaly data, and DeFi-specific exploit databases to enrich the feature set with signals more directly tied to crypto infrastructure. Shorter sub-annual prediction intervals would increase the number of labeled observations available for training. More rigorous causal analysis – using Granger causality tests or instrumental variable approaches exploiting exogenous macroeconomic shocks – could move the pipeline from correlation to mechanism. Extending the dataset to cover non-English-language sources and incidents below public reporting thresholds would reduce selection bias and increase coverage of the full global threat landscape.

## References


[1] A. Olushola and S. P. Meenakshi, "Cybersecurity crimes in cryptocurrency exchanges (2009–2024)," *Frontiers in Blockchain*, vol. 8, p. 1713637, 2025.

[2] Center for Governance of Technology and Systems, *Cyber Events Database*, University of Maryland, 2026. [Online]. Available: https://gotech.umd.edu/cyber-events-database

[3] Center for Strategic and International Studies, *Significant Cyber Incidents Since 2006*, 2026. [Online]. Available: https://www.csis.org/programs/strategic-technologies-program/significant-cyber-incidents

[4] U.S. Department of Health and Human Services / Office for Civil Rights, *Breach Portal: Notice to the Secretary of HHS*, 2026. [Online]. Available: https://ocrportal.hhs.gov/ocr/breach/breach_report.jsf

[5] Washington State Office of the Attorney General, *Data Breach Notifications Affecting Washington Residents*, 2026. [Online]. Available: https://www.atg.wa.gov/data-breach-notifications

[6] J. Platt, "Probabilistic outputs for support vector machines and comparisons to regularised likelihood methods," in *Advances in Large Margin Classifiers*, MIT Press, 1999, pp. 61–74.

[7] R. J. Hyndman and G. Athanasopoulos, *Forecasting: Principles and Practice*, 3rd ed. OTexts, 2021. [Online]. Available: https://otexts.com/fpp3/

[8] N. Atzei, M. Bartoletti, and T. Cimoli, "A survey of attacks on Ethereum smart contracts," in *Proc. 6th Int. Conf. Principles of Security and Trust (POST)*, Lecture Notes in Computer Science, vol. 10204, Springer, 2017, pp. 164–186.

[9] S. Romanosky, "Examining the costs and causes of cyber incidents," *Journal of Cybersecurity*, vol. 2, no. 2, pp. 121–135, 2016.

[10] Kaggle / Information is Beautiful, *World's Biggest Data Breaches & Hacks*, 2022. [Online]. Available: https://www.informationisbeautiful.net/visualizations/worlds-biggest-data-breaches-hacks/